

Report for Challenge Zoomer

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ABSTRACT

This is a report on our solution of the Zoomer Challenge in the PINE 26 seminar. We were given an executable used within an authentication system. The challenge required analyzing the executable, recovering the embedded machine learning model and reconstructing the hidden 18 training samples.

The scoring used for this challenge was mean cosine similarity across all hidden samples. Our final solution is a natural image retrieval approach combined with ablation as an image retrieval strategy. Our final solution achieved a semantic similarity score of 0.6657.

1 INITIAL ANALYSIS

Before we proceeded with our development approach for reconstructing the hidden training images, we tried to reconstruct the preprocessing pipeline, identify the model architecture and recover the weights used by the executable. This allowed us to reconstruct the executable in a white-box scenario, which gave us access to inner features, intermediate tensor values and allowed us to evaluate large batches of candidate images.

1.1 Reverse engineering the model

We managed to reverse engineer the executable in a PyTorch white-box model. By using probe .ras black and white images as input to the executable, and by inspecting the executable dynamically using the GNU Debugger, we observed that the tensor values were identical to the values obtained after applying Z-score normalization with the mean and standard deviation of ImageNet [1] after applying [0, 1] scaling. We observed that the normalization was applied to every channel.

Afterwards, by inspecting and disassembling a helper function named `get_weights(const char *)`, we managed to find that the model uses a table named `weight_index` and a contiguous `float32` blob. The table leaked parameter names, offsets, size counts for each tensor and the location of the blob. Tensor names such as `conv`, `bn`, `fc` and layer numbers from 1 – 4 combined with two basic blocks per layer and the preprocessing used, indicated that the executable used a ResNet-18 [2] architecture. We also observed that the final classification layer was different from standard ResNet-18. The executable used a `Linear(512, 1)` binary head. The table, combined with the blob location, allowed us to extract the model weights and fully reconstruct it using PyTorch.

2 INITIAL APPROACH

Our initial approach is natural image retrieval from public datasets. We chose our approach on the assumption that the model is trained using publicly available images.

2.1 Concept

Before we applied the natural image retrieval approach, we needed to construct a retrieval pool of publicly available images to select

samples. The pool consists of images from datasets such as COCO, Visual Genome, Places365 and Open Images V7 [4–6, 11]. We used our white-box model reconstruction to process batches of images and score the samples. Afterwards, we filtered our image pool to contain only samples with positive logit values. Samples with the highest positive values in their final logits were selected into sets of 18 candidate images for evaluation by the remote challenge server, as we believed that the reconstructed model associated them strongly with the authenticated class.

In parallel, considering that the hidden set has 18 images, we ran a diverse selection branch to prevent submission of near duplicate images due to having a high similarity with only one of the hidden training images. For diversity-based selection, we selected images with high positive logits but with low mean cosine similarity to the already selected images. The candidate images are represented by normalized image features ϕ_i and classifier logits ℓ_i . We first selected the image with the maximum logit. Each next image was chosen greedily by

$$i^* = \arg \min_{i \notin S} \left[\max_{j \in S} \phi_i^\top \phi_j - \lambda \ell_i \right],$$

where S is the current selected set. We used $\lambda = 0$ for pure diversity based selection, $\lambda = 0.05$ for and $\lambda = 0.10$ to account for high positive logit values as a weight in the selection. As another measure of diversity, we selected cluster representative images by running k -means with $k = 18$ in the same feature space and selected the highest-logit image from each cluster.

2.2 Implementation

We started by retrieving the images from our dataset pool via `retrieve_natural_candidates.py` (lines 58 – 65). The model was loaded via the function `load_model()` and the preprocessing was applied via `tensor_from_crops()` function. We obtained the logits from our reconstructed classifier at the `main()` function. The candidate selection strategy was applied at `select_retrieval_variants.py` via the functions `select_top()`, `select_diverse()` and `select_clusters()`. The submission of candidates to the remote challenge server for scoring was done via `batch_remote_submission.py`.

2.3 Evaluation

The top logits retrieval strategy achieved a similarity score of 0.562. The diversification strategy achieved a similar similarity score of 0.5614. The low scores were attributed to the broadness of our retrieval pool and various image themes being selected as candidates.

3 IMPROVED VERSION

The motivation behind this approach was the need to perform a fine-grained search to identify themes in the highest-scoring images and select samples from those themes. This was based on the assumption that the hidden training set consists of one specific theme.

3.1 Concept

Our new approach was to search for images via textual descriptions of the target theme. It was inspired by Radford et al. [8], who in their work managed to map images and text into a shared embedding space, allowing us to compare the similarity of an image to a given text prompt. Using a pre-trained CLIP [8] encoder, we embedded a text label to every image on top of the logits embedded by the classifier. This allowed us to observe the textual embeddings that received a high mean cosine similarity score, identify the relevant theme and use textual prompts from that theme to refine our search for candidates that have a high mean cosine similarity to the relevant text prompt of the theme in the shared embedding space.

The work from Sultan et al. [9] supports our approach and demonstrates that using CLIP with natural language descriptions allowed for more targeted queries. To provide a baseline for comparison, we used DINOv2 [7] to extract feature vectors and to search for images with similar feature vectors, using only image-to-image similarity to retrieve new candidates.

3.2 Implementation

After observing the images with the highest scores, we created a prompt file called `prompts.txt`, with keywords we assumed would reach a high semantic similarity. At `scene_prior_retrieval.py`, we used the prompts to filter the image search, attached embeddings to sample images and to textual prompts via CLIP [8] using the `clip_prefilter()` function. Prompt similarity scores are computed at line 204. We implemented the DINOv2 [7] embedding generation and embedding extraction for both variants at `semantic_retrieval_variants.py` using the `featurize_transformer()` function. The embeddings are later used inside the same script for creating candidate sets at lines 171-223.

3.3 Evaluation

The semantic retrieval approach achieved a semantic similarity score of 0.6613, using CLIP [8] semantic embeddings. This approach achieved an improvement compared to the previous iteration. The DINOv2 [7] embeddings achieved a best score of 0.5744 on the COCO [6] dataset. The failure of DINOv2 was attributed to the model’s inability to capture high-level semantic meaning between different images.

4 FINAL SUBMISSION

The motivation behind our final approach was that during the previous iteration, we found the relevant semantic region, but we had yet to pick the optimal 18 candidates from our semantic region. We treated the problem as a local-search optimization problem,

Table 1: Comparison of the main versions using the hidden mean cosine similarity score.

Version	Mean cosine similarity
Initial approach	0.5620
Improved version	0.6613
Final version	0.6657

trying to pick the highest-scoring candidate in every slot from our retrieval pool.

4.1 Concept

Our approach consists of anchoring the set obtained from the previous iteration. The concept is similar to the work from Hoos and Stützle [3] which improves a combinatorial solution through small neighbourhood moves, in our case, the combinatorial solution being the ranked 18-image set of candidates and the small neighbourhood move being the candidate swap. We calculated a proxy score to determine which candidates are weak and to evaluate replacement candidates. The proxy score combined similarity to the current best 18-image anchor, similarity to the semantic prompt set, and the normalized reconstructed classifier logit.

$$s(x) = \text{anchor-sim}(x) + \alpha \text{ theme-sim}(x) + \beta \text{ logit}(x)$$

With α and β being adjustable weights, in our approach, we set them to $\alpha = 0.25$ and $\beta = 0.05$, to prioritize similarity to the anchor set centre, which is the average vector of all CLIP embeddings of the 18 anchor images. The theme similarity is the maximum mean cosine similarity score between one of the textual embeddings listed in the prompt file and the candidate image. Based on this proxy score, we ranked weak candidates in the anchor set and ordered replacement images using the proxy score.

Afterwards, we generated swap combinations between a weak candidate and potential candidates high in the list (strong fillers) and candidates in the middle of the list (neutral fillers). The neutral fillers were included due to the fact that a high proxy score did not always translate to a higher similarity score. The remote challenge server was used as an oracle to test whether the swap improved the similarity score. This process was done iteratively until the similarity score reached a plateau.

4.2 Implementation

We calculated the proxy score at `generate_ablation_tests.py` (lines 233-238). The candidate ranking and swap combination generation is at the `main()` function (lines 246-289). We generated swaps between weak slots and top replacements at `generate_single_swap_tests.py` (lines 270-289).

4.3 Evaluation

This approach achieved the highest score of 0.6657. During the iterations of this approach, we noticed a plateau in the similarity score. We believe that we achieved the optimal set of candidates for this retrieval pool, and to achieve a higher similarity score, we would need to include images from other sources.

5 DISCUSSION

Our solution reached a similarity score of 0.6657, demonstrating that the natural image retrieval approach could lead to a baseline similarity score in this hidden image reconstruction challenge. We focused on a natural image retrieval approach, improving the retrieval of images similar to the hidden set and lastly, we optimized our candidate selection. Our approach was limited by the number of publicly available image datasets and hypothetically, if the classifier were trained using private images, our approach would fail to

fully reconstruct the hidden image set. The natural image approach is still useful to find relevant clues and themes about the hidden images.

Due to time constraints, we were unable to fully pursue synthetic image generation as a viable alternative approach to the challenge problem. In future work, we recommend pursuing alternative approaches such as synthetic image generation based on work from Yin et al. [10].

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